stable (lower energy), intermediate:

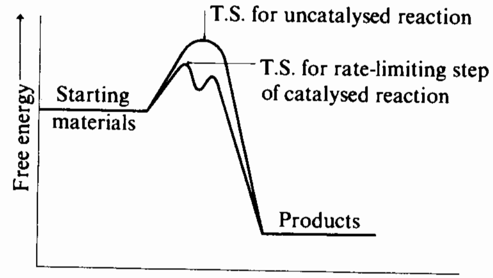
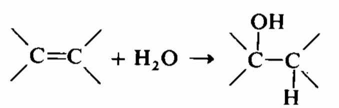
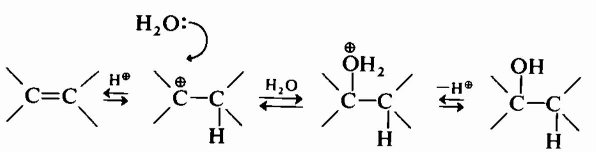


Fig. 2.5

Thus the rate of hydration of an alkene, directly with water,



is often extremely slow, but it can be greatly speeded up by the presence of an acid catalyst, which effects initial protonation of the alkene to a carbocationic intermediate. This is then followed by easy and rapid attack on the now positively charged carbocation by a water molecule acting as a nucleophile, and finally by liberation of a proton which is able to function again as a catalyst (p. 187):



The details of acid/base catalysis are discussed subsequently (p. 74).

2.2.3 Kinetic versus thermodynamic control

Where a starting material may be converted into two or more alternative products, e.g. in electrophilic attack on an aromatic species that already carries a substituent (p. 150), the proportions in which the alternative products are formed are often determined by their relative rate of formation : the faster a product is formed the more of it there will be in the final product mixture; this is known as *kinetic control.* This is not always what is observed however, for if one or more of the

alternative reactions is reversible, or if the products are readily interconvertible directly under the conditions of the reaction, the composition of the final product mixture may be dictated not by the relative rates of formation of the different products, but by their relative thermodynamic stabilities in the reaction system: we are then seeing *thermodynamic* or *equilibrium control.* Thus the nitration of methylbenzene is found to be kinetically controlled, whereas the Friedel–Crafts alkylation of the same species is often thermodynamically controlled (p. 163). The form of control that operates may also be influenced by the reaction condition, thus the sulphonation of naphthalene with concentrated H_2SO_4 at 80° is essentially kinetically controlled, whereas at 160° it is thermodynamically controlled (p. 164).

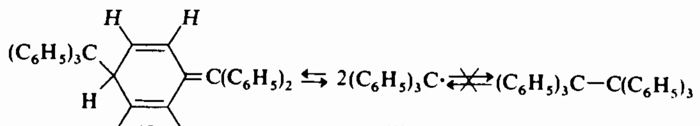
2.3 INVESTIGATION OF REACTION MECHANISMS

It is seldom, if ever, possible to provide complete and entire information, structural, energetic, and stereochemical, about the pathway that is traversed by any chemical reaction: no reaction mechanism can ever be proved to be correct! Sufficient data can nevertheless usually be gathered to show that one or more theoretically possible mechanisms are just not compatible with the experimental results, and/or to demonstrate that of several remaining alternatives one is a good deal more likely than the others.

2.3.1 The nature of the products

Perhaps the most fundamental information about a reaction is provided by establishing the structure of the products that are formed during its course, and relating this information to the structure of the starting material. Where, as is often the case with organic reactions, more than one product is obtained then it is usually an advantage to know also the relative proportions in which the products are obtained, e.g. in establishing, among other things, whether kinetic or thermodynamic control is operating (*cf.* p. 42). In the past this had to be done laboriously—and often imprecisely—by manual isolation of the products, but may now often be achieved more easily, and precisely, by sophisticated chromatographic methods or, indirectly, by suitable spectroscopic ones.

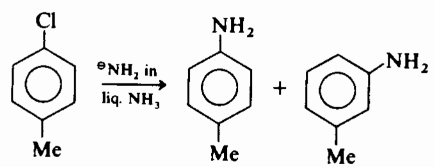
The importance of establishing the correct structure of the reaction product is best illustrated by the confusion that can result when this has been assumed, wrongly, as self-evident, or established erroneously. Thus the yellow triphenylmethyl radical (3, *cf.* p. 300), obtained from the action of silver on triphenylmethyl chloride in 1900, readily forms a colourless dimer (m.w. = 486) which was—reasonably enough—assumed to be hexaphenylethane (4) with *thirty* 'aromatic' hydrogen atoms. Only after nearly seventy years (in 1968) did the n.m.r. spectrum (*cf.* p. 18) of the dimer (with only *twenty-five* 'aromatic' (H), *four* 'dienic' (*H*), and *one* 'saturated' (**H**), hydrogen atoms) demonstrate that it could not have the hexaphenylethane structure (4) and was, in fact (5):



(5) (3) (4)

At which point numerous small details of the behaviour of (3) and of its dimer, that had previously appeared anomalous, promptly became understandable.

Information about the products of a reaction can be particularly informative when one of them is quite unexpected. Thus the reaction of chloro-4-methylbenzene (*p*-chlorotoluene, 6) with amide ion, $^{\ominus}NH_2$, in liquid ammonia (p. 173) is found to lead not only to the expected 4-methylphenylamine (*p*-toluidine, 7), but also to the quite unexpected 3-methylphenylamine (*m*-toluidine, 8), which is in fact the major product:



(6) (7) (8)

Expected Unexpected

The latter clearly cannot be obtained from (6) by a simple substitution process, and either must be formed from (6) *via* a different pathway than (7), or if the two products are formed through some common intermediate then clearly (7) cannot be formed by a direct substitution either.

2.3.2 Kinetic data

The largest body of information about reaction pathways has come—and still does come— from kinetic studies as we shall see, but the interpretation of kinetic data in mechanistic terms (*cf.* p. 39) is not always quite as simple as might at first sight be supposed. Thus the effective reacting species, whose concentration really determines the reaction rate, may differ from the species that was put into the reaction mixture to start with, and whose changing concentration we are actually seeking to measure. Thus in aromatic nitration the effective

attacking species is usually $^{\oplus}NO_2$ (p. 134), but it is HNO_3 that we put into the reaction mixture, and whose changing concentration we are measuring; the relationship between the two may well be complex and so, therefore, may be the relation between the rate of reaction and $[HNO_3]$. Despite the fact that the essential reaction is a simple one, it may not be easy to deduce this from the quantities that we can readily measure.

Then again, if the hydrolysis in aqueous solution of the alkyl halide, RHal, is found to follow the rate equation,

$$\text{Rate} = k_1[\text{RHal}]$$

it is not necessarily safe to conclude that the rate-determining step does not involve the participation of water, simply on the grounds that $[H_2O]$ does not appear in the rate equation; for if water is being used as the solvent it will be present in very large excess, and its concentration would remain virtually unchanged whether or not it actually participated in the rate-limiting stage. The point could perhaps be settled by carrying out the hydrolysis in another solvent, e.g. HCO_2H, and by using a much smaller concentration of water as a potential nucleophile. The hydrolysis may then be found to follow the rate equation,

$$\text{Rate} = k_2[\text{RHal}][H_2O]$$

but the actual mechanism of hydrolysis could well have changed on altering the solvent, so that we are not, of necessity, any the wiser about what actually went on in the original aqueous solution.

The vast majority of organic reactions are carried out in solution, and quite small changes in the solvent used can have the profoundest effects on reaction rates and mechanisms. Particularly is this so when polar intermediates, for example carbocations or carbanions as constituents of ion pairs, are involved, for such species normally carry an envelope of solvent molecules about with them. This greatly affects their stability (and their ease of formation), and is strongly influenced by the composition and nature of the solvent employed, particularly its polarity and ion-solvating capabilities. By contrast, reactions that involve radicals (p. 299) are much less influenced by the nature of the solvent (unless this is itself capable of reacting with radicals), but are greatly influenced by the addition of radical sources (e.g. peroxides) or radical absorbers (e.g. quinones), or by light which may initiate reaction through the production of radicals by photochemical activation, e.g. $Br_2 \xrightarrow{h\nu} Br\cdot \ \cdot Br$.

A reaction that is found, on kinetic investigation, to proceed unexpectedly faster or slower than the apparently similar reactions, under comparable conditions, of compounds of related structure suggests the operation of a different, or modified, pathway from the general one that might otherwise have been assumed for the series. Thus the observed rates of hydrolysis of the chloromethanes with strong bases are found, under comparable conditions, to vary as follows,

$$CH_3Cl \gg CH_2Cl_2 \ll CHCl_3 \gg CCl_4$$

clearly suggesting that trichloromethane undergoes hydrolysis in a different manner from the other compounds (*cf.* p. 267).

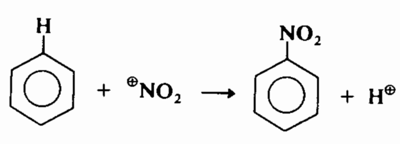
2.3.3 The use of isotopes

It is often a matter of some concern to know whether a particular bond has, or has not, been broken in a step up to and including the rate-limiting step of a reaction: simple kinetic data cannot tell us this, and further refinements have to be resorted to. If, for example, the bond concerned is C—H, the question may be settled by comparing the rates of reaction, under the same conditions, of the compound in which we are interested, and its exact analogue in which this bond has been replaced by a C—D linkage. The two bonds will have the same chemical nature as isotopes of the same element are involved, but their vibration frequencies, and hence their dissociation energies, will be slightly different because atoms of different *mass* are involved: the greater the mass, the stronger the bond. This difference in bond strength will, of course, be reflected in different rates of breaking of the two bonds under comparable conditions: the weaker C—H bond being broken more rapidly than the stronger C—D bond; quantum-mechanical calculation suggests a maximum rate difference, k_H/k_D, of ≈ 7 at 25°.

Thus in the oxidation

$$Ph_2C(OH)H \xrightarrow[^{\ominus}OH]{MnO_4^{\ominus}} Ph_2C{=}O$$

it is found that Ph_2CHOH is oxidised 6·7 times as rapidly as Ph_2CDOH; the reaction is said to exhibit a *primary kinetic isotope effect*, and breaking of the C—H bond must clearly be involved in the rate-limiting step of the reaction. By contrast benzene, C_6H_6, and hexadeuterobenzene, C_6D_6, are found to undergo nitration at essentially the same rate, and C—H bond-breaking, that must occur at some stage in the overall process,



thus cannot be involved in the rate-limiting step (*cf.* p. 136).

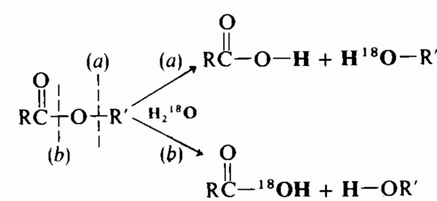
Primary kinetic isotope effects are also observable with pairs of isotopes other than hydrogen/deuterium, but as the relative mass difference must needs be smaller their maximum values will be correspondingly smaller. Thus the following have been observed:

$$HO^{\ominus} + {}^{12}_{(14)}CH_3{-}I \longrightarrow HO{-}{}^{12}_{(14)}CH_3 + I^{\ominus} \qquad \frac{k_{^{12}C}}{k_{^{14}C}} = 1{\cdot}09\ (25°)$$

$$PhCH_2{-}{}^{35}_{(37)}Cl + H_2O \longrightarrow PhCH_2{-}OH + H^{\oplus}\,{}^{35}_{(37)}Cl^{\ominus} \qquad \frac{k_{^{35}Cl}}{k_{^{37}Cl}} = 1{\cdot}0076\ (25°)$$

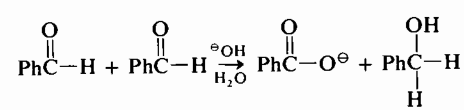
It should be emphasised that primary kinetic isotope effects are observed experimentally with values intermediate between the maximum calculated value and unity (i.e. *no* isotope effect): these too can be useful, as they may supply important information about the breaking of particular bonds in the transition state.

Isotopes can also be used to solve mechanistic problems that are non-kinetic. Thus the aqueous hydrolysis of esters to yield an acid and an alcohol could, in theory, proceed by cleavage at (*a*) alkyl/oxygen fission, or (*b*) acyl/oxygen fission:



If the reaction is carried out in water enriched in the heavier oxygen isotope ^{18}O, (*a*) will lead to an alcohol which is ^{18}O enriched and an acid which is not, while (*b*) will lead to an ^{18}O enriched acid but a normal alcohol. Most simple esters are in fact found to yield an ^{18}O enriched acid indicating that hydrolysis, under these conditions, proceeds *via* (*b*) acyl/oxygen fission (p. 238). It should of course be emphasised that these results are only valid provided that neither acid nor alcohol, once formed, can itself exchange its oxygen with water enriched in ^{18}O, as has indeed been shown to be the case.

Heavy water, D_2O, has often been used in a rather similar way. Thus in the Cannizzaro reaction of benzaldehyde (p. 216),



(9)

the question arises of whether the second hydrogen atom that becomes attached to carbon, in the molecule of phenylmethanol (benzyl alcohol,

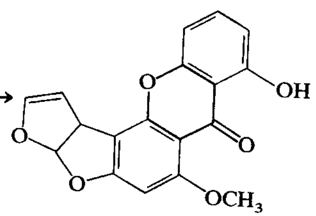
9) that is formed, comes from the solvent (H_2O) or from a second molecule of benzaldehyde. Carrying out the reaction in D_2O is found to lead to the formation of no PhCHDOH, thus demonstrating that the second hydrogen atom could not have come from water, and must therefore have been provided by direct transfer from a second molecule of benzaldehyde.

A wide range of other isotopic labels, e.g. ^3H (or T), ^{13}C, ^{14}C, ^{15}N, ^{32}P, ^{35}S, ^{37}Cl, ^{131}I, etc., have also been used to provide important mechanistic information. The major difficulties encountered in such labelling studies have always been: (*a*) ensuring that the label is incorporated *only* into the desired position(s) in the test compound; and (*b*) finding exactly where the label has gone to in the product(s) after the reaction being studied has taken place.

The enormous increase in selectivity of modern synthetic methods has all but eliminated (*a*), but (*b*) long remained a major problem; particularly when isotopes of carbon were being used: these being especially valuable because carbon atoms are present in all organic compounds. The ^{14}C isotope has been much used in investigating biosynthetic pathways: the routes by which living organisms build up the highly elaborate molecules that may be obtained from them.

Thus there was reason to believe that the pentacyclic compound *sterigmatocystin*,

(i) $^{14}CH_3CO_2H$
(ii) $CH_3{}^{14}CO_2H$ ⟶



(i) 8 ^{14}**C** atoms
(ii) 9 ^{14}**C** atoms

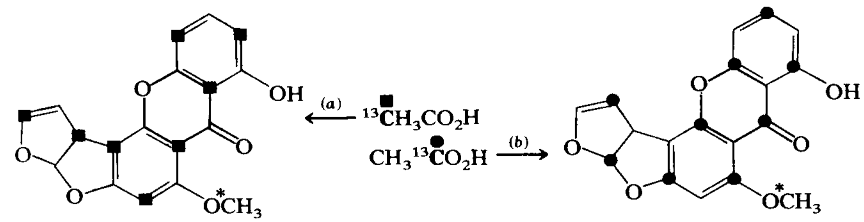
sterigmatocystin

found in cultures of several fungi, was built up stepwise from molecules of ethanoic acid. General confirmation of this hypothesis was obtained through feeding suitable fungal cultures, in separate experiments, with $^{14}CH_3CO_2H$ and $CH_3{}^{14}CO_2H$, respectively. It was then found from radioactive counting measurements (^{14}C is a β emitter), on the two extracted samples of sterigmatocystin ($C_{18}H_{12}O_6$) that: (i) $^{14}CH_3CO_2H$ led to the introduction of 8^{14}**C** atoms, and $CH_3{}^{14}CO_2H$ to the introduction of 9^{14}**C** atoms. But that still leaves open the question of exactly where in the sterigmatocystin molecule these two sets of labelled carbon atoms are located.

Not long ago this could have been determined only by extremely laborious, and often equivocal, selective degradation experiments; but the coming of carbon n.m.r. spectroscopy has now made all the difference. Neither the ^{12}C nor the ^{14}C carbon isotopes produce an n.m.r. signal but the ^{13}C isotope, which occurs in ordinary carbon to

the extent of 1·11%, does. It is thus possible, with suitable instrumentation, to record ^{13}C n.m.r. spectra of all carbon-containing compounds (because of their 1·11% ^{13}C content): each carbon atom, or group of identically situated carbon atoms, in a molecule producing a distinguishably different signal.

The ^{13}C spectrum of normal sterigmatocystin can thus be compared with the spectra of the molecules resulting from separate feeding experiments with ^{13}C enriched (*a*) ^{13}**C**H_3CO_2H and (*b*) $CH_3{}^{13}$**C**O_2H, respectively. Those carbon atoms, in each case, which now show enhanced ^{13}C signals can thereby be identified:



(*a*) 8 ^{13}**C**■ atoms (*b*) 9 ^{13}**C**● atoms

Knowing which of the two carbon atoms in CH_3CO_2H molecules are incorporated into which positions in sterigmatocystin, it becomes possible to make pertinent suggestions about the synthetic pathway employed by the fungal cultures. Incidentally, it also shows that the methyl carbon of the $^*CH_3O$ group does *not* come from CH_3CO_2H.

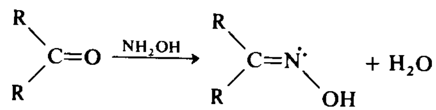
2.3.4 The study of intermediates

Among the most concrete evidence obtainable about the mechanism of a reaction is that provided by the actual isolation of one or more intermediates from the reaction mixture. Thus in the Hofmann reaction (p. 122), by which amides are converted into amines,

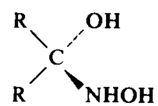
$$RC(=O){-}NH_2 \xrightarrow[\ominus OH]{Br_2} RNH_2$$

it is, with care, possible to isolate the N-bromoamide, RCONHBr, its anion, $RCONBr^{\ominus}$, and an isocyanate, RNCO; thus going some considerable way to elucidate the overall mechanism of the reaction. It is of course necessary to establish beyond all doubt that any species isolated really is an intermediate—and not merely an alternative product—by showing that it may be converted, under the normal reaction conditions, into the usual reaction products at a rate at least as fast as the overall reaction under the same conditions. It is also important to establish that the species isolated really is on the direct reaction pathway, and not merely in equilibrium with the true intermediate.

It is much more common not to be able to isolate any intermediates at all, but this does not necessarily mean that none are formed, merely that they may be too labile or transient to permit of their isolation. Their occurrence may then often be inferred from physical, particularly spectroscopic, measurements made on the system. Thus in the formation of oximes from a number of carbonyl compounds by reaction with hydroxylamine (p. 219),



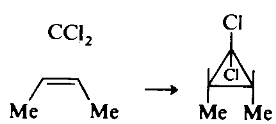
the infra-red absorption band characteristic of C=O in the starting material disappears rapidly, and may have gone completely before the band characteristic of C=N in the product even begins to appear. Clearly an intermediate must be formed, and further evidence suggests that it is the *carbinolamine* (10),



(10)

which forms rapidly and then breaks down only slowly to yield the products, the oxime and water.

Where we have reason to suspect the involvement of a particular species as a labile intermediate in the course of a reaction, it may be possible to confirm our suspicions by introducing into the reaction mixture, with malice aforethought, a reactive species which we should expect our postulated intermediate to react with particularly readily. It may then be possible to divert the labile intermediate from the main reaction pathway—to *trap* it—and to isolate a stable species into which it has been unequivocally incorporated. Thus in the hydrolysis of trichloromethane with strong bases (*cf.* p. 46), the highly electron-deficient *dichlorocarbene*, CCl_2, which has been suggested as a labile intermediate (p. 267), was 'trapped' by introducing into the reaction mixture the electron-rich species *cis* but-2-ene (11), and then isolating the resultant stable cyclopropane derivative (12), whose formation can hardly be accounted for in any other way:



(11) (12)

The successful study of intermediates not only provides one or more signposts which help define the detailed pathway traversed by a reaction, the intermediates themselves may also provide inferential evidence about the transition states for which they are often taken as models (*cf.* p. 41).

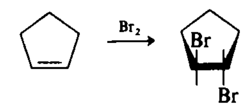
2.3.5 Stereochemical criteria

Information about the stereochemical course followed by a particular reaction can also provide useful insight into its mechanism, and may well introduce stringent criteria that any suggested mechanistic scheme will have to meet. Thus the fact that the base-catalysed bromination of an optically active stereoisomer of the ketone (13)

$$\underset{(+)}{\mathrm{PhCO\overset{*}{C}HMeEt}} \xrightarrow[\ominus\mathrm{OH}]{\mathrm{Br_2}} \underset{(\pm)}{\mathrm{PhCO\overset{*}{C}BrMeEt}}$$

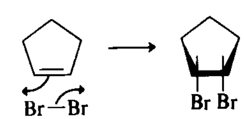
(13)

leads to an optically inactive racemic product (p. 295), indicates that the reaction must proceed through a planar intermediate, which can undergo attack equally well from either side leading to equal amounts of the two mirror-image forms of the product. Then again, the fact that cyclopentene (14) adds on bromine under polar conditions to yield the *trans* dibromide (15) only, indicates that the mechanism of



(14) (15)

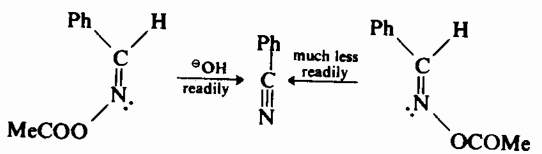
the reaction cannot simply be direct, one-step addition of the bromine molecule to the double bond, for this must lead to the *cis* dibromide (16):



(14) (16)

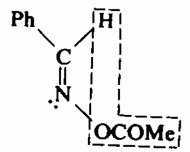
The addition must be at least a two-step process (*cf.* p. 179). Reactions like this, which proceed so as to give largely—or even wholly—one stereoisomer out of the two alternatives possible, are said to be *stereoselective.*

Then again, many elimination reactions are found to occur much more readily in that member of a pair of geometrical isomerides in which the atoms or groups to be eliminated are *trans* to each other, than in the isomer in which they are *cis* (p. 255). As is seen in the relative ease of elimination from *anti* and *syn* aldoxime acetates to yield the same cyanide:



Anti *Syn*

This clearly sets limitations to which any mechanism advanced for the reaction will have to conform, and gives the lie to that prime tenet of 'lasso chemistry': groups are eliminated most readily when closest together:



The degree of success with which a suggested mechanism can be said to delineate the course of a particular reaction is not determined solely by its ability to account for the known facts; the acid test is how successful it is at forecasting a change in rate, or even in the nature of the products formed, when the conditions under which the reaction is carried out, or the structure of the starting material, are changed. Some of the suggested mechanisms we shall encounter measure up to these criteria better than do others, but the overall success of a mechanistic approach to organic reactions is demonstrated by the way in which the application of a few relatively simple guiding principles can bring light and order to bear on a vast mass of disparate information about equilibria, reaction rates, and the relative reactivity of organic compounds. We shall now go on to consider some simple examples of this.

3

The strengths of acids and bases

Modern electronic theories of organic chemistry have been highly successful in a wide variety of fields in correlating behaviour with structure, not least in accounting for the relative strengths of organic acids and bases. According to the definition of Arrhenius, acids are compounds that yield hydrogen ions, $H^{\oplus}$, in solution while bases yield hydroxide ions, $^{\ominus}OH$. Such definitions are reasonably adequate if reactions in water only are to be considered, but the acid/base relationship has proved so useful in practice that the concepts of both acids and bases have become considerably more generalised. Thus Brønsted defined acids as substances that would give up protons, i.e. *proton donors*, while bases were *proton acceptors*. The first ionisation of sulphuric acid in aqueous solution is then looked upon as:

$$\underset{\text{Acid}}{H_2SO_4} + \underset{\text{Base}}{H_2O:} \rightleftarrows \underset{\text{Conjugate acid}}{H_3O^{\oplus}} + \underset{\text{Conjugate base}}{HSO_4^{\ominus}}$$

Here water is acting as a base by accepting a proton, and is thereby converted into its so-called *conjugate acid*, $H_3O^{\oplus}$, while the acid, H_2SO_4, by donating a proton is converted into its *conjugate base*, $HSO_4^{\ominus}$.